Koushik Mondal

Software Engineer — Satellite Simulation & Scalable Web Systems

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SUMMARY

Software Engineer with 3+ years building high-performance satellite simulation and geospatial web systems. Specialized in Next.js, React, TypeScript, and real-time 3D visualization with Cesium JS. Proven record of delivering scalable, production-grade frontend architectures in the aerospace sector, with deep expertise in performance optimization and clean system design.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript, HTML5, CSS3
Frontend	React.js, Next.js, Tailwind CSS, ShadCN UI, Aceternity UI, Bootstrap
3D / Geospatial	Cesium JS, Mapbox
Backend	Node.js, Express.js, REST API, GraphQL, tRPC, oRPC
Databases & ORM	PostgreSQL, MongoDB, Prisma ORM
State / Fetching	Redux, Zustand, Context API, TanStack Query
Cloud & Tools	Supabase, Convex, Azure DevOps, Git, GitHub, Jira

PROFESSIONAL EXPERIENCE

Software Engineer
Feb 2023 – Present

Antaris Space India Pvt Ltd (ASIL) · Pune, India

- Architected satellite simulation interfaces for designing satellites with payloads, bus, and edge systems.
- Built real-time 3D visualization with Cesium JS for interactive orbit simulation and dual-mode tracking (real & virtual).
- Implemented end-to-end pre- and post-simulation workflows supporting the full mission lifecycle in production.
- Designed scalable Next.js frontend architecture (App Router, SSR, code splitting) for high-concurrency aerospace workloads.

Software Engineer
Nov 2022 – Present

Geminus Tech · Pune, India

- Developed scalable React + TypeScript frontend applications with production-quality codebases and maintainable architecture.
- Integrated REST and GraphQL APIs using TanStack Query with optimized caching strategies for performance-critical features.
- Built reusable UI component libraries accelerating team velocity and enforcing design system consistency across products.

KEY PROJECTS

Global Intelligence Monitoring Platform (World Monitor)

- Engineered a real-time global data visualization platform handling large-scale geospatial layers with dynamic Mapbox-based interfaces.
- Optimized rendering pipelines and data ingestion to support enterprise-scale monitoring with smooth interactivity at thousands of data points.

Satellite Simulation Dashboard (Antaris Simulation)

- Delivered a full-featured simulation platform for configuring satellites with modular hardware components and complete lifecycle tracking.
- Built advanced 3D Cesium visualizations with custom camera controls, orbit rendering, and interactive simulation overlays.

CORE COMPETENCIES

- Distributed System Design & Scalable Architecture
- Real-time 3D Visualization & Geospatial Systems
- Performance Optimization & Core Web Vitals
- API Design (REST, GraphQL, tRPC)
- Clean Code & Maintainability Best Practices
- Cross-functional Collaboration & Fast Iteration